Lab 9: Pre Lab Questions

Start Assignment

- Due Tuesday by 5pm
- Points 15
- Submitting a file upload

Lab 9: Pre Lab Questions

Individual assignment. You need the Lab 9 page and the CMV300 datasheet. No board, no Vivado.

Question 2 is the prediction that Milestone 3 is measured against. Bring your numbers to the session.

Question 1: The SPI frame (5 points)

The frame is 16 bits, MSB first: one control bit, seven address bits, eight data bits.

a. Give, in hex, the 16-bit word that reads register 83, and the word that writes the value 251 to register 83.

b. The sensor samples SPI_IN on the rising edge of SPI_CLK and launches SPI_OUT on the falling edge. State which edge your master should drive SPI_IN on, and which edge it should sample SPI_OUT on. Justify both from the sensor's behaviour, not from convention.

c. The register defaults include reg 1 = 232 and reg 2 = 1, the low and high bytes of Number_lines. Reconstruct it, and say what physical property of the sensor that number is. Then: if your shift register were off by one bit on reads, what would you compute instead, and would you notice?

Question 2: Derive the timing, in both regimes (5 points)

CLK_IN is 25 MHz. From the datasheet: a row is 324 pixels, one invalid slot, then 324 pixels, with LVAL low for one clock between rows, so budget **650 CLK_IN periods per row**. The frame overhead time with reg 58 at its default is **4875 clocks**.

a. Compute, in milliseconds: one row, the FOT, the readout of a 488-row frame, and the exposure the default Exp_time gives.

b. **One frame, from idle.** The sensor exposes, then does the FOT, then reads out, in that order. How long is the silence between your FRAME_REQ pulse and the first LVAL, and how long from the first LVAL to the last?

c. **Pipelined.** With Number_frames greater than 1 the next exposure overlaps the current readout, so the frame period is FOT + max(readout, exposure). Compute it at the default exposure, give the ratio to your answer in (b), and explain in one or two sentences what pipelining actually saves.

d. Sketch pipelined frame period against exposure from 1 ms to 40 ms. At what exposure is the knee, and what Exp_time value is that?

> Milestone 3 measures every one of these. A prediction you got wrong and can explain is worth full marks; a prediction you did not make is worth nothing, because there is then nothing for the measurement to disagree with.

Question 3: The preconditions and the errata (5 points)

a. The PLL bands are 251 for 10–32 MHz, 187 for 20–50 MHz and 155 for 32–50 MHz. The power-on default for reg 83 is 155 and your CLK_IN is 25 MHz. Which value should it be, and what does the datasheet say? With the wrong band the PLL still locks and the CLK_OUT meter reads about 36,700 cycles per millisecond instead of 25,000 — by what percentage would every timing measurement then be wrong?

b. List the five preconditions from the Lab 9 page that must all hold before FRAME_REQ does anything. For each, state the one counter reading that would reveal it was not met.

c. Two of the five are timing requirements measured in microseconds. Explain in one or two sentences why they must be enforced by hardware in your sequencer rather than by a sleep in Python. Then: FRAME_REQ must be a pulse, not a level. Describe what a held-high FRAME_REQ produces, and which of the two regimes from question 2 it puts the sensor into.